

Tableau Practical Lesson 4 Submission

Performing Data Analysis using Calculations in Tableau

A. Basic Calculated Field

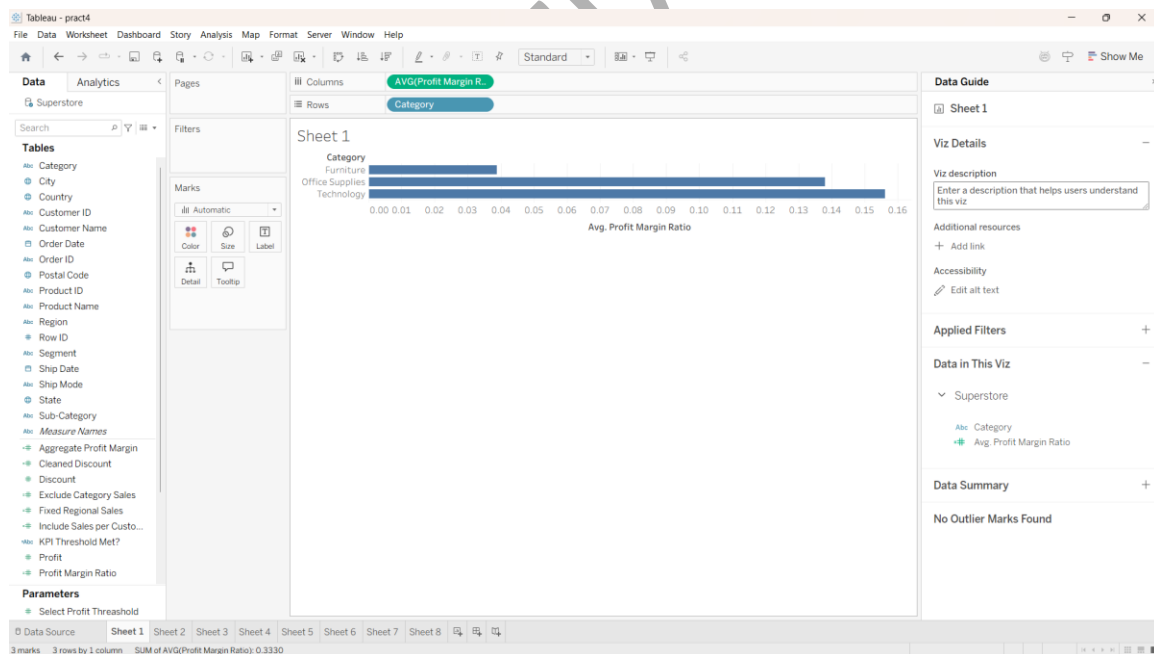
Formula: $[\text{Profit}]/[\text{Sales}]$

Objective: Create a Profit Margin Ratio to measure profitability instead of only sales.

Procedure:

1. Create Calculated Field named Profit Margin Ratio.
2. Enter formula.
3. Drag Category to Rows.
4. Drag Profit Margin Ratio to Columns.
5. Change aggregation to Average.

Observation: Categories are compared by average profit margin.



B. Row-Level Calculation

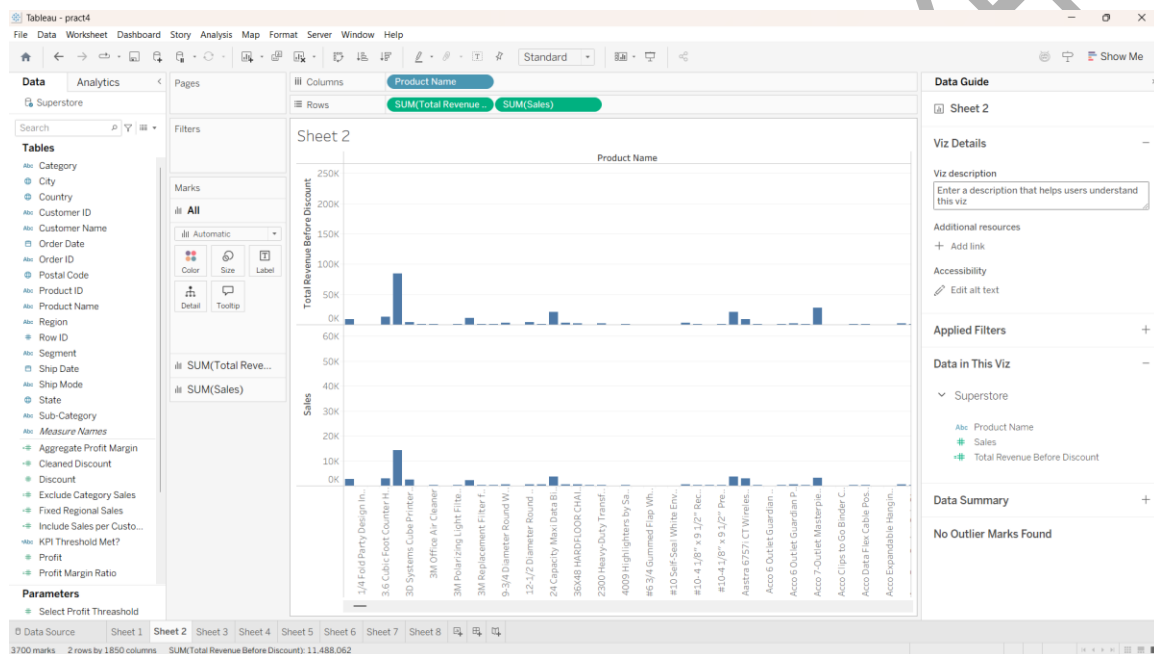
Formula: $[\text{Sales}] * [\text{Quantity}]$

Objective: Calculate revenue for each transaction.

Procedure:

1. Create calculated field.
2. Drag Product Name to Rows.
3. Drag Total Revenue Before Discount to Columns.
4. Add Sales for comparison.

Observation: Calculation is performed for every row.



C. Aggregate Calculation

Formula: $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$

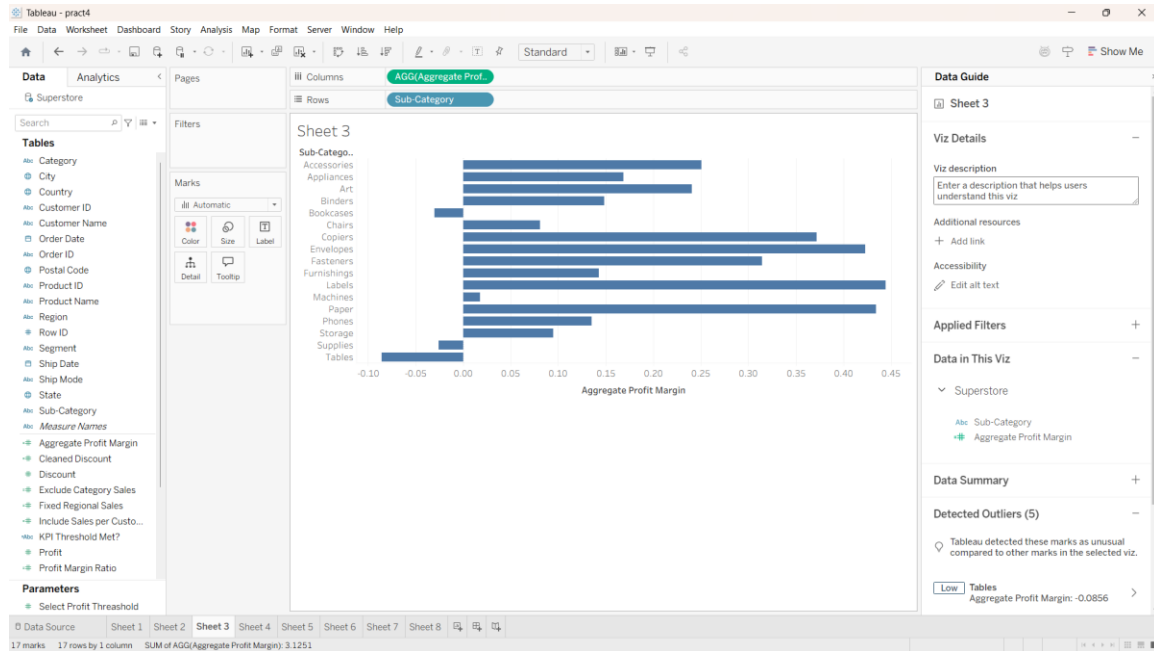
Objective: Calculate correct overall profit margin.

Procedure:

1. Create Aggregate Profit Margin.
2. Drag Sub-Category to Rows.

3. Drag Aggregate Profit Margin to Columns.

Observation: Displays AGG field.



D. FIXED LOD

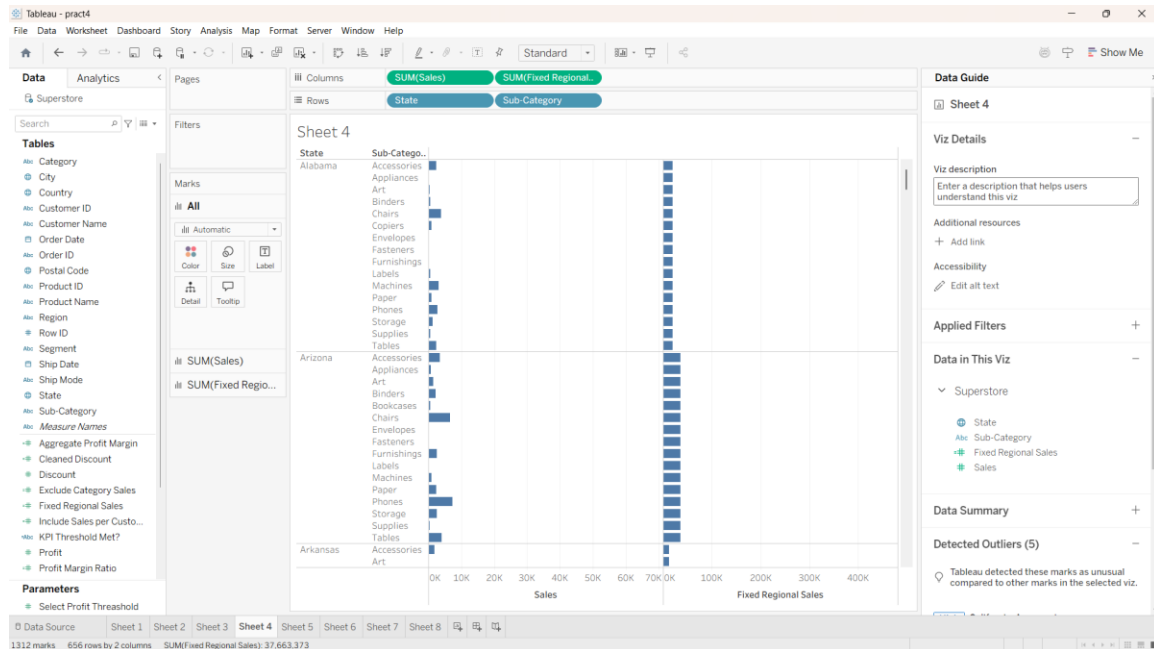
Formula: $\{ \text{FIXED } [\text{State}] : \text{SUM}([\text{Sales}]) \}$

Objective: Lock sales total at State level.

Procedure:

1. Create Fixed Regional Sales.
2. Rows: State then Sub-Category.
3. Columns: Sales and Fixed Regional Sales.

Observation: State total repeats for every sub-category.



E. INCLUDE LOD

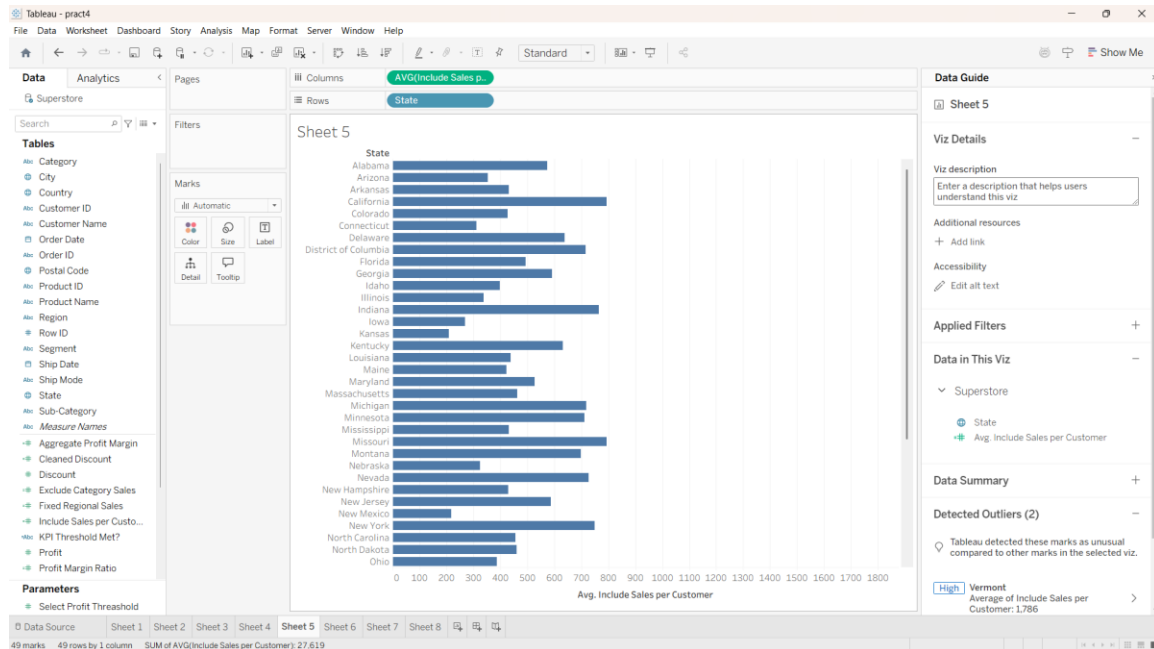
Formula: { INCLUDE [Customer ID] : SUM([Sales]) }

Objective: Calculate customer-level sales while viewing state.

Procedure:

1. Create field.
2. Rows: State.
3. Columns: Include Sales per Customer.
4. Change to Average.

Observation: Average customer sales per state.



F. EXCLUDE LOD

Formula: { EXCLUDE [Sub-Category] : SUM([Sales]) }

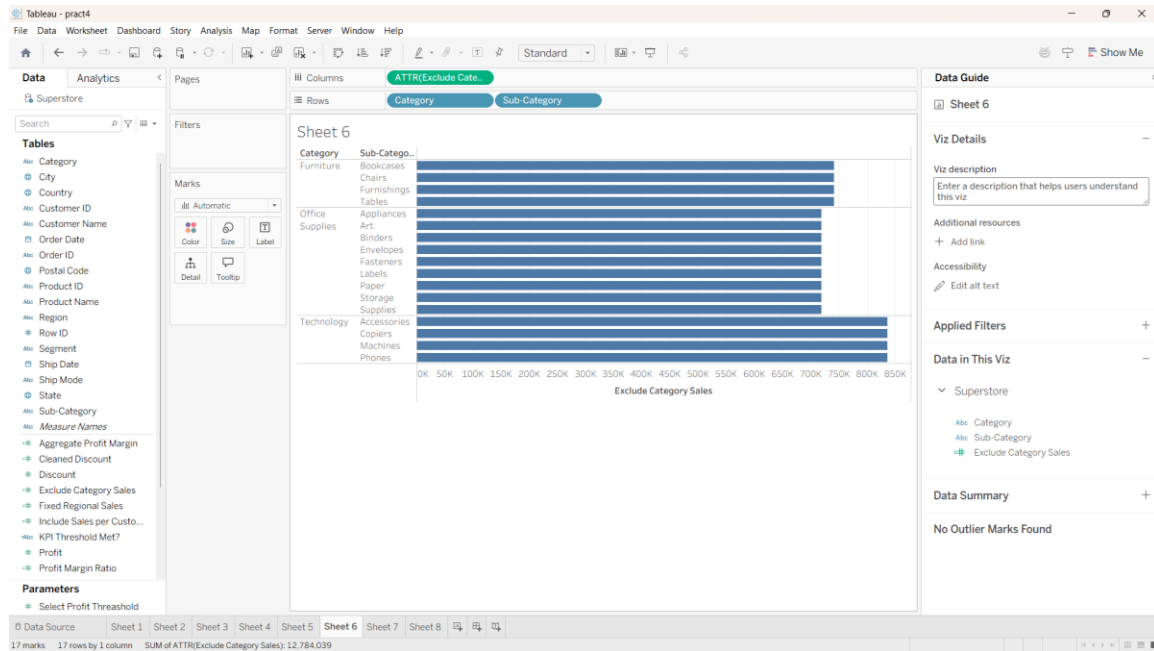
Objective: Ignore Sub-Category during calculation.

Procedure:

1. Rows: Category and Sub-Category.

2. Columns: Exclude Category Sales.

Observation: Category total repeats.



G. Parameter

Parameter: Select Profit Threshold

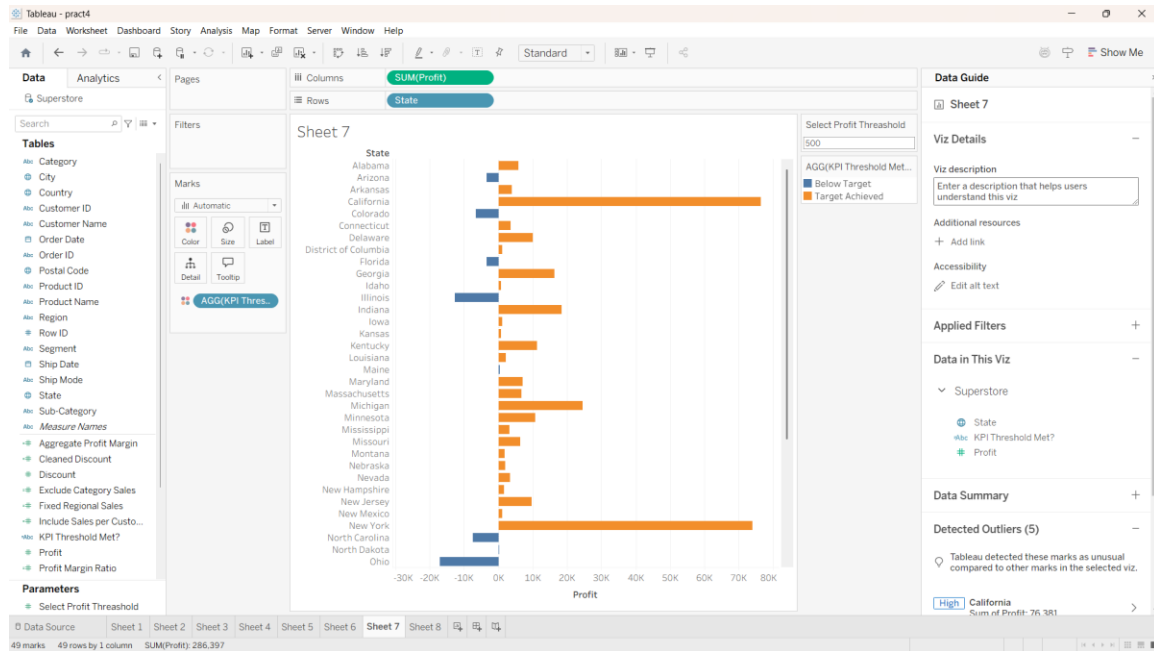
Calculated Field:

IF SUM([Profit]) >= [Select Profit Threshold] THEN 'Target Achieved' ELSE 'Below Target' END

Procedure:

1. Create parameter.
2. Create calculated field.
3. Rows: State.
4. Columns: Profit.
5. Color: KPI Threshold Met?
6. Show Parameter Control.

Observation: Colors change with threshold.



H. Data Cleaning

IFNULL([Discount],0)

UPPER([City Type])

Objective: Replace nulls and standardize text.

Procedure:

1. Create both calculated fields.
2. Rows: Standardized City Type.
3. Columns: Cleaned Discount (Average).

Observation: Nulls replaced and text standardized.

